X(3872) Analysis Update

From ψ(2S) reweighting to the current PbPb strategy

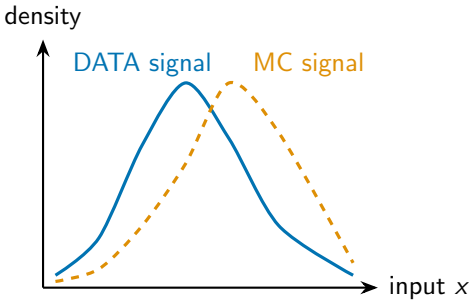
Lekai Yao

CMS

2026.7.29

ppRef prototype completed — PbPb selection and fit diagnostics in progress

Why can MC mismodelling bias the X analysis?



The classifier sees the simulated signal distribution, not the unknown true X signal distribution.

Signal MC $\neq$ signal DATA in reconstructed features

↓

XGBoost learns simulation-specific structure

↓

Decision boundary shifts when applied to DATA

↓

Selection efficiency and mass-fit stability become uncertain

A more powerful classifier does not by itself remove a domain mismatch.

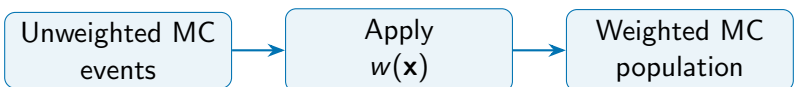
Event reweighting as multidimensional density-ratio estimation

Target correction

$$w(\mathbf{x}) \simeq \frac{p_{\text{DATA signal}}(\mathbf{x})}{p_{\text{MC signal}}(\mathbf{x})}.$$

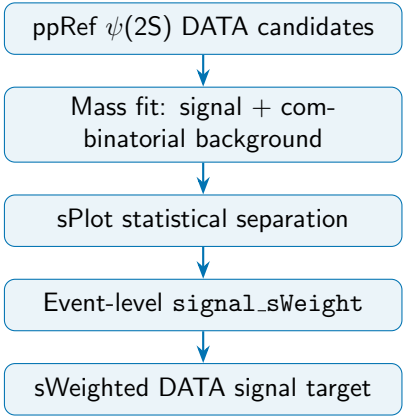
$$p_{\text{MC}}^{\text{weighted}}(\mathbf{x}) \approx p_{\text{DATA signal}}(\mathbf{x}).$$

- $\mathbf{x}$: correlated reconstruction-level variables
- One continuous weight per simulated event
- BDT reweighter handles the multidimensional ratio
- No event reconstruction is changed
- No new simulated events are created



BDT reweighting: A. Rogozhnikov, arXiv:1608.05806.

The DATA target must represent signal, not inclusive background



What the target is

- Statistical representation of the signal component
- Negative sWeights retained
- Fit and effective-statistics diagnostics required

What it is not

- Not an event-by-event truth label
- Not equivalent to a pure simulated signal sample
- Not usable without a trustworthy mass fit

What is transferred from $\psi(2S)$ to X?

Transfer

- DATA/MC correction in shared variables
- Tracking and vertex reconstruction response
- Correction inside validated phase-space support
- Event weights applied to X simulation

Do not transfer

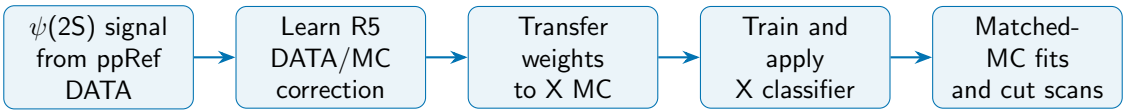
- Raw $\psi(2S)$ physics distribution
- $\psi(2S)$ mass or dipion dynamics
- A $\psi(2S)$-trained classifier
- Unconstrained correction outside target support

Transfer assumption to validate

$$\left[\frac{p_{\mathrm{DATA}}}{p_{\mathrm{MC}}}\right]_{\psi(2S)}(\mathbf{x} \mid \mathbf{z}) \approx \left[\frac{p_{\mathrm{DATA}}}{p_{\mathrm{MC}}}\right]_{X}(\mathbf{x} \mid \mathbf{z}),$$

$\mathbf{x}$: shared detector/reconstruction variables; $\mathbf{z}$: conditioning kinematics such as p_{T} and rapidity.

Analysis path: validate the chain, then move to PbPb



ppRef question

Can the complete correction $\rightarrow$ ML $\rightarrow$ fit workflow be validated?

Interpretation boundary

Observed significance alone does not establish the final training preference.

The ppRef prototype is now closed; the next bottleneck is PbPb ψ(2S).

Reweighting profiles: variable definitions

Profile	Reweighting variables
R3	`Bcos_dtheta`, `Btktkpt`, `Bchi2Prob`
R4-noCos	`Btktkpt`, `Bchi2Prob`, `Btrk1Pt`, `Btrk2Pt`
R5	`Bcos_dtheta`, `Btktkpt`, `Bchi2Prob`, `Btrk1Pt`, `Btrk2Pt`
R8	`Bchi2Prob`, `Btrk1dR`, `BtrkPtimb`, `Btrk1Pt`, `Btrk2Pt`, `BtktkvProb`, `Bcos_dtheta`, `Btktkpt`

Controlled comparisons

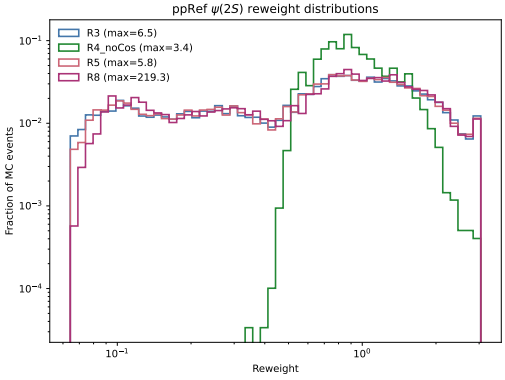
R3: low-dimensional check; R4-noCos: remove `Bcos_dtheta`.

Prototype decision

R5 nominal; R8 excluded because of weight tails and loss of effective statistics.

Reweighting profiles: KS-like closure metric

Profile	Mean before	Mean after
R3	0.0908	0.0100
R4-noCos	0.0908	0.0409
R5	0.0908	**0.0091**
R8	0.0908	0.0123



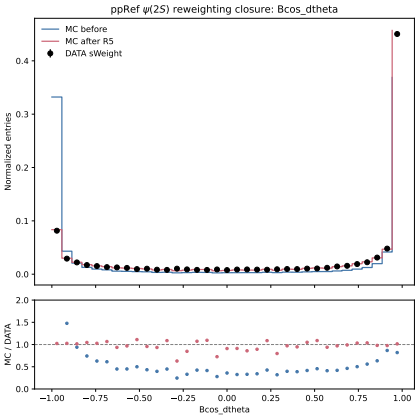
Metric interpretation

Mean maximum CDF distance over eight validation variables, using signed target weights. Descriptive KS-like distance, not a KS p-value.

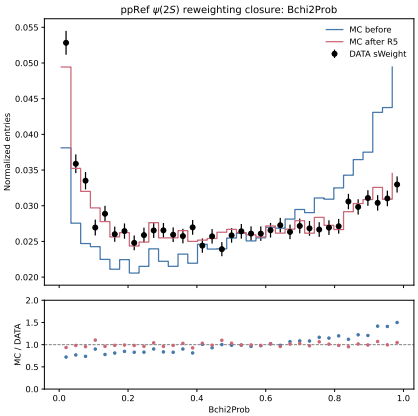
R5 gives the best average one-dimensional closure without the R8 weight instability.

One-dimensional closure: before and after R5

Bcos_dtheta

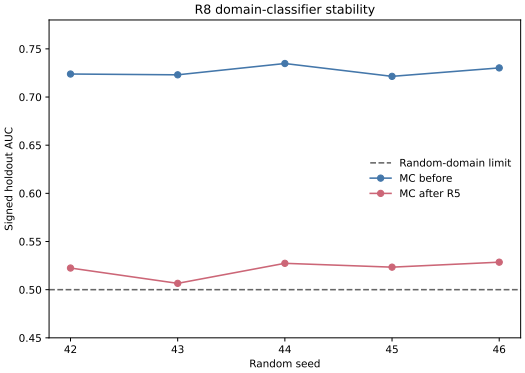


Bchi2Prob



One-dimensional agreement is necessary, but correlations require an independent multivariate test.

Closure: little residual DATA/MC separation



Independent holdout test

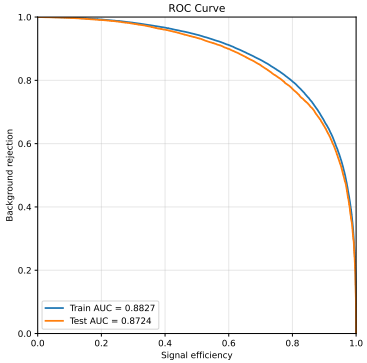
Domain-classifier AUC $\simeq 0.5225$, stable over five random splits.

- Weight statistics validated
- X MC support/coverage checked
- `Reweight` written to X MC
- No obvious residual separation

R5 passes the ppRef closure and transfer checks used in this prototype.

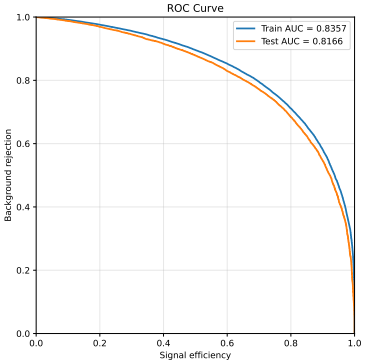
X classifier: unweighted and R5-weighted 8v2

Unweighted X MC



`X_pp24_v4_fid3_8v2_rw0_xgb_v1`

R5-weighted X MC



`X_pp24_v4_fid3_8v2_rwpsi2sr5v1_xgb_v1`

Both train/apply/draw chains and scored DATA/MC ROOT files are complete.

Matched-MC working points

Training	Bkg. eff.	Status	Yield	Signif.	Mass [GeV]
rw0	10%	0/0	762 ± 63	12.18	3.87028
rw0	3%	$-1/3$	286 ± 80	8.04	3.86884
rw0	1%	0/0	112 ± 20	5.52	3.86703
R5	10%	0/0	775 ± 65	12.21	3.87174
R5	3%	0/0	216 ± 54	5.84	3.87229
R5	1%	0/0	78 ± 23	3.40	3.87208

Interface fixed

Each training uses its own scored DATA and matched MC template.

Comparison caveat

rw0 3% did not converge; it cannot support a quantitative ranking.

No consistent R5 sensitivity gain across the three predefined working points.

Punzi optimum: useful diagnostic, not a final ranking

Training	Frozen cut	Efficiency	Fit status	Significance
rw0	0.875	0.2840	strategy-2: 0/0	7.01
R5	0.613	0.6775	0/0	16.62

What was validated

- Frozen before inspecting new fits
- R5 fit converges with a clear X peak
- R5 cut agrees with the later 0.6 scan point

Why this is not the answer

- Efficiency definitions differ
- rw0 cut 0.875 is too tight in DATA
- Significance cannot rank the schemes alone

Fixed-cut scans expose the relevant trend

Scan	Prediction range	Significance trend	Fitted mass range [GeV]
rw0	0.2–0.8	$20.27 \rightarrow 9.49$	$3.87128 \rightarrow 3.86914$
R5	0.2–0.8	$\sim 20 \rightarrow 6.63$	3.87113–3.87204

rw0

- Seven default fits converge
- Yield/significance fall with tighter cuts
- Frozen 0.875 is beyond the useful scan

R5

- More stable fitted peak position
- Cut 0.8 passes strategy-2 check
- $S(0.613) = 16.62$, $S(0.6) = 16.91$

ppRef validates the chain, but does not uniquely choose the final correction scheme.

PbPb $\psi(2S)$: cumulative S0–S4 definitions

ID	Selection
S0	`BQvalue < 0.15 && abs(By) < 1.6 && Bpt > 10`
S1	S0, but tighten `abs(By) < 1.2`
S2	S1 `&& Btrk2dR < 0.25`
S3	S2 `&& BtrkPtimb > 0.15`
S4	S3 `&& Btrk1dR < 0.25`

Design

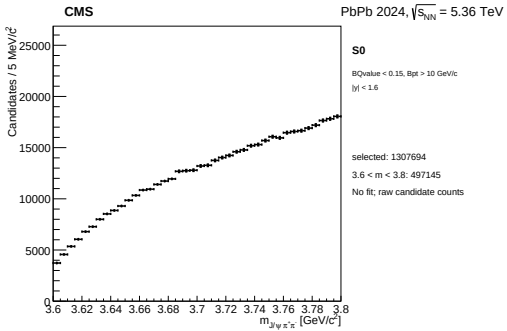
- Same $3.6 < m < 3.8$ GeV range
- Same 5 MeV binning
- Strictly cumulative selections

ML interaction

- `Btrk2dR`: not planned for R5 ML
- `BtrkPtimb`, `Btrk1dR`: ML candidates
- S3/S4 are diagnostic, not nominal

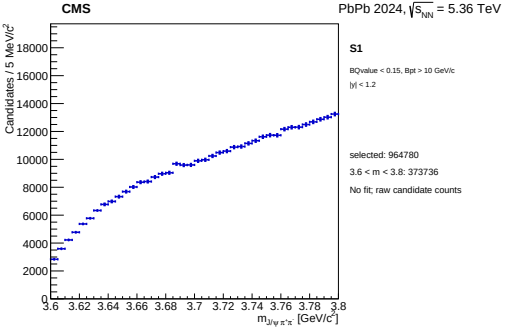
PbPb ψ(2S) mass spectra: S0 and S1

S0 — common fiducial baseline



497,145 entries in $3.6 < m < 3.8$ GeV

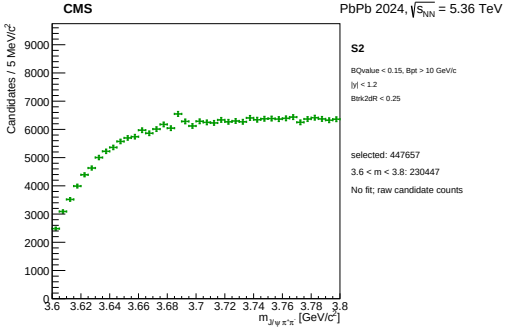
S1 — tighter rapidity



373,736 entries in $3.6 < m < 3.8$ GeV

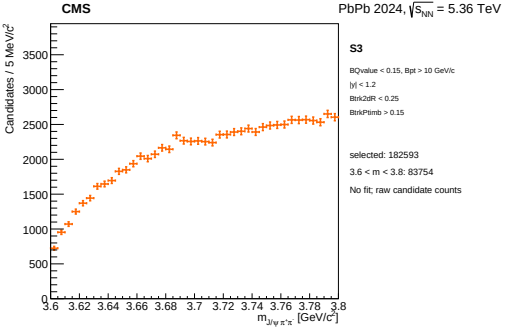
PbPb ψ(2S) mass spectra: S2 and S3

S2 — add `Btrk2dR < 0.25`



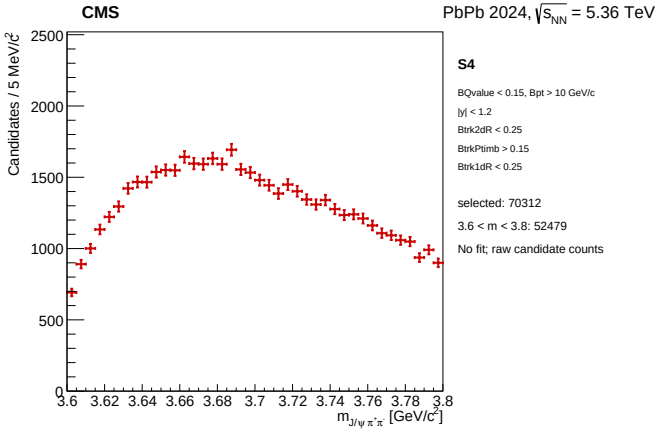
230,447 entries; preferred fit candidate

S3 — add `BtrkPtimb > 0.15`



83,754 entries; tighter fallback

PbPb $\psi(2S)$ mass spectrum: S4



S4: strict diagnostic point

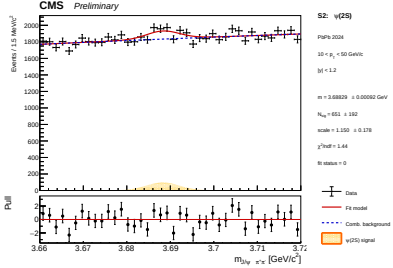
- Add `Btrk1dR < 0.25`
- 52,479 entries in the main window
- Cuts two ML-candidate variables

Decision

S2 and S3, not S4, were retained for the consistent preliminary fits.

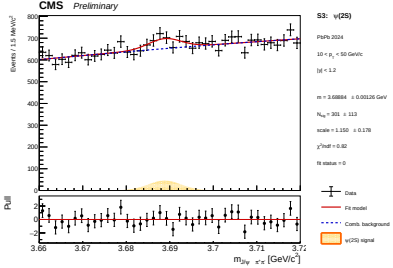
Current checkpoint: preliminary PbPb S2/S3 fits

S2 — preferred candidate



$$N_{\psi(2S)} = 651 \pm 192, \qquad \frac{S}{\sqrt{S+B}} = 4.39$$

S3 — tighter fallback



$$N_{\psi(2S)} = 301 \pm 113, \qquad \frac{S}{\sqrt{S+B}} = 3.41$$

Fit diagnostics in progress

Both fits have status/covQual $= 0/3$; the DATA width scale reaches 1.15. Results are preliminary; sWeight and N_{eff} checks are pending.

Status and immediate next steps

Completed: ppRef prototype

- ψ(2S) signal target and R5 correction
- Closure and transfer to X MC
- rw0/R5 train, apply, matched fits
- Prediction scans and physical review

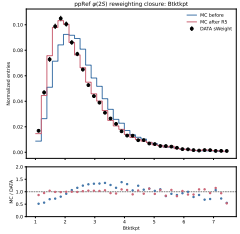
Next: PbPb-specific chain

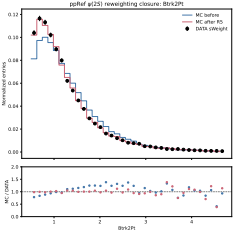
1. Accept S2 or S3 fit configuration
2. Produce and validate sWeights
3. Train PbPb-specific correction
4. Retrain X ML and scan Prediction

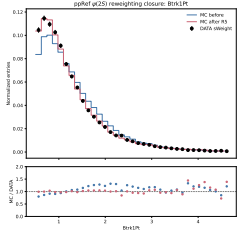
The ppRef chain is validated.

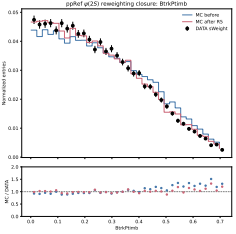
The immediate decision is whether the preliminary PbPb ψ(2S) fit is robust enough to provide the next reweighting target.

Backup: additional R5 variable closure









Backup: rw0 fixed-cut scan

Cut	DATA	MC	Status	Yield	Signif.	Mass [GeV]
0.2	104673	82611	0/0	2995 ± 156	20.27	3.87128
0.3	84800	79852	0/0	2596 ± 168	19.56	3.87098
0.4	67465	76051	0/0	2045 ± 156	18.15	3.87075
0.5	51473	70655	0/0	1647 ± 133	16.98	3.87069
0.6	35775	62777	0/0	1141 ± 88	14.56	3.87040
0.7	22553	52784	0/0	741 ± 63	11.92	3.87026
0.8	10684	38329	0/0	419 ± 59	9.49	3.86914

All seven default fits converge; the frozen Punzi cut 0.875 lies beyond this scan.

Backup: R5 fixed-cut scan

Cut	DATA	MC	Status	Yield	Signif.	Mass [GeV]
0.2	126742	83853	0/0	3319 ± 200	20.49	3.87142
0.3	104199	81419	0/0	2903 ± 169	19.71	3.87132
0.4	83658	77565	0/0	2637 ± 170	19.92	3.87124
0.5	64331	71343	0/0	1969 ± 153	17.87	3.87113
0.6	46432	62052	0/0	1581 ± 130	16.91	3.87131
0.7	27656	46252	0/0	960 ± 113	13.86	3.87186
0.8	9026	19775	0/0*	256 ± 40	6.63	3.87204

*Cut 0.8 converges after the predefined strategy-2 stability check.

Backup: PbPb $\psi(2S)$ selection statistics

Selection	Total	$3.6 < m < 3.8$
S0	1,307,694	497,145
S1	964,780	373,736
S2	447,657	230,447
S3	182,593	83,754
S4	70,312	52,479

Selection scan

- Flat PbPb24 DATA: 25,496,039 entries
- Mass spectra only; no automatic optimization
- S2/S3 chosen after visual inspection

Preliminary fit diagnostics

- S2 preferred; S3 tighter fallback
- Width scale reaches upper bound 1.15
- sWeight diagnostics still pending